

FALL RIVER HARBOR, MA, USA

WMO: 997279

Lat: 41.7043N

Lon: 71.1640W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.7	(c) 16.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.3	(c) 87.6	(d) N/A	(e) 84.9	(f) N/A	(g) 82.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	31.2	32.8	39.2	50.9	59.9	68.5	76.2	73.7	67.4	56.6	46.4	39.1		
	DBStd	16.71	8.94	7.80	7.90	6.10	6.59	6.02	4.22	4.12	5.79	6.50	7.75	8.40		
	HDD50	2004	584	481	346	60	3	0	0	0	0	18	160	352		
	HDD65	5066	1048	901	799	424	187	33	1	1	40	269	559	804		
	CDD50	3321	1	0	11	88	309	555	814	736	523	222	50	12		
	CDD65	910	0	0	0	2	28	138	349	272	113	8	0	0		
	CDH74	5054	0	0	0	23	135	672	2340	1458	408	18	0	0		
	CDH80	1151	0	0	0	4	22	143	642	278	61	1	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	47.0	3.8	3.5	4.7	4.2	3.5	3.7	3.4	3.8	4.0	4.1	4.0	4.4		
	PrecMax	57.4	6.3	7.1	14.7	8.0	7.8	11.3	9.1	6.7	9.2	12.8	6.5	8.0		
	PrecMin	36.1	1.7	1.8	0.2	1.5	1.2	0.1	1.1	0.8	0.7	0.3	0.7	1.3		
	PrecStd	6.0	1.2	1.4	2.9	1.7	1.4	2.9	1.6	1.7	2.0	2.6	1.5	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.4	54.9	66.0	77.7	83.1	88.7	92.1	89.5	86.1	76.4	67.4	62.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	55.0	50.6	58.1	67.6	77.8	84.3	88.6	85.3	81.1	72.0	63.4	58.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	49.5	47.2	54.0	63.4	74.2	80.8	86.1	83.2	78.2	68.7	60.6	55.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	44.5	43.9	50.3	60.4	70.3	77.9	83.6	81.0	76.0	66.2	58.3	52.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		12.4	12.7	13.2	14.9	14.9	14.0	13.3	13.0	13.8	12.3	12.4	11.5		
	5% DB	MCDBR	18.6	16.4	17.6	18.9	19.7	17.2	15.4	14.7	14.2	14.3	14.0	14.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.310	0.334	0.372	0.412	0.460	0.484	0.439	0.374	0.341	0.313	0.301		
	taud		2.506	2.487	2.466	2.388	2.311	2.215	2.156	2.271	2.429	2.530	2.566	2.561		
	Ebn at Noon		272	287	291	285	275	261	252	261	273	272	266	263		
	Edh at Noon		23	27	31	36	40	45	47	41	32	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		584	862	1183	1523	1692	1840	1901	1708	1388	937	639	489		
	RadStd		44	82	86	163	168	179	103	88	100	83	46	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	+85

Nomenclature: See separate page